

tions and zero for low-exposure occupations; γ_{it} are firm-by-time fixed effects; and $\xi_{p(i)ct}$ are industry-by-exposure-by-time fixed effects. Equation 12 has the same structure as the seniority-based triple-difference in Equation 11, but here the third dimension is *occupational exposure* (high vs. low) rather than seniority. The coefficients β_j^s trace, separately for juniors and seniors, how employment in high-exposure occupations evolves relative to low-exposure occupations within the same firm and quarter, comparing adopters to non-adopters.

Figure 6 plots the estimated β_j^s coefficients for juniors and seniors, with both series normalized to zero in 2022Q4, the last pre-ChatGPT quarter. Two patterns stand out. First, throughout the pre-ChatGPT period, high-exposure occupations expand faster than low-exposure occupations at adopters relative to non-adopters for both seniority groups, consistent with adopters tilting their workforce toward exposed occupations over time. The two profiles track each other closely through 2022, with the senior coefficients rising somewhat faster than the junior coefficients. Second, beginning in 2023Q1, the two series diverge sharply. The junior series declines steadily, reaching a roughly 7 log point decline in high-exposure versus low-exposure junior employment at adopters by 2025Q1, relative to the 2022Q4 baseline. In contrast, the senior series continues to rise after 2022Q4, with no visible break around the ChatGPT launch.

Taken together, these results provide two insights. First, the post-2022 contraction in junior employment at adopters is concentrated in high-exposure occupations rather than reflecting a broad-based decline across all junior roles. Second, the design helps rule out a purely compositional explanation—namely, that high-exposure occupations declined economy-wide after 2023 and adopters simply employed a larger share of such roles. By comparing high- and low-exposure occupations *within* the same firm-by-quarter, and by running the same specification for seniors as a placebo-like comparison, we find that high-exposure occupations contract relative to low-exposure occupations only for juniors and only at adopters after the ChatGPT launch. The senior profile continues along its pre-existing upward trend, consistent with the interpretation that GenAI adoption is seniority-biased: it shifts demand away from junior workers in the very occupations where the technology is most applicable, while leaving senior employment in those same occupations essentially unaffected.

Supplemental Appendix A.3 reports a complementary heterogeneity exercise that re-

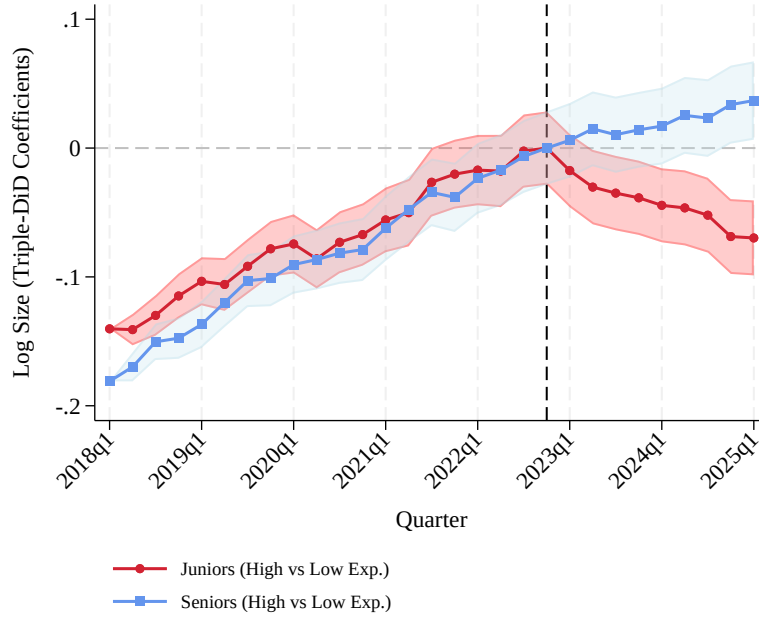


Figure 6: Triple-Difference by Occupational Exposure—Juniors vs. Seniors

Notes: This figure reports the estimated coefficients β_j^s from Equation 12, estimated separately for juniors (red) and seniors (blue). Each series captures the evolution of log employment in high-exposure occupations relative to low-exposure occupations at adopters relative to non-adopters, conditional on firm-by-time and industry-by-exposure-by-time fixed effects. Details on the exposure measure are provided in Section 4.1. Shaded bands are 95% confidence intervals. Standard errors are clustered at the firm level.

estimates the DiD specification within tiers of juniors' educational background, measured by the prestige of their alma mater. The relative decline in junior employment is U-shaped across tiers: largest for juniors from mid-tier institutions and smallest for juniors from the most and least selective institutions.

5.2.4 Discussion: The Timing of the Junior Employment Decline

The early and pronounced decline in the triple-difference coefficients—starting to emerge in 2023Q1, shortly after the release of ChatGPT—may appear surprising, given that automation effects typically unfold more gradually. One plausible explanation is that the rapid diffusion of GenAI tools at the end of 2022 constituted a discrete shock to firms' expectations about future automation. In response, firms may have adjusted in a forward-looking manner by scaling back hiring for junior roles they anticipated would be automated in the near future. Such preemptive adjustments could be optimal for firms if

they view slower hiring as less costly than future layoffs.²⁹ Supplemental Appendix A.5 formalizes this mechanism in a simple dynamic model in which expectations of *future* automation—combined with labor-adjustment costs—lead firms to reduce hiring *today*.

One might still question whether firms were sufficiently aware of GenAI tools by early 2023 for such expectation shifts to occur. However, the sharp surge in adoption from 2023Q1 onward—as shown in Figure 2 and documented by studies such as Bick et al. (2024)—suggests that firms’ behavior changed almost immediately following the release of ChatGPT. Consistent with this, mentions of “AI” in firms’ earnings calls began rising steeply as early as 2022Q4, tripling by 2023Q2 (Appendix A.20). Finally, Supplemental Appendix A.21 compiles anecdotal evidence indicating that several prominent U.S. companies publicly adjusted their hiring practices and internal workflows in early 2023, explicitly attributing these changes to the adoption or anticipated impact of GenAI.

5.3 Staggered Event-Study of Junior Employment Around Adoption

We complement the triple-difference analyses with a staggered event study that traces junior employment dynamics around the timing of GenAI adoption, proxied by the first period in which a firm posts a GenAI integrator vacancy. This design helps distinguish adoption effects from broader time-specific shocks by exploiting variation in adoption timing across firms. However, it is sensitive to measurement error in the adoption proxy—for example, if firms begin using GenAI before posting for an integrator role or only several periods afterward. Specifically, we estimate:

$$\log(\text{JuniorEmployment}_{it}) = \alpha + \sum_{j=2}^J \beta_j (\text{Lag}_j)_{it} + \sum_{k=1}^K \gamma_k (\text{Lead}_k)_{it} + \mu_i + \lambda_t + \varepsilon_{it}, \quad (13)$$

where $\log(\text{JuniorEmployment}_{it})$ denotes the log number of junior workers at firm i at time t ; $(\text{Lag}_j)_{it}$ is an indicator equal to one if the current period t is j periods before adoption; and $(\text{Lead}_k)_{it}$ is defined analogously for periods after adoption. μ_i and λ_t are firm and time fixed effects, and ε_{it} is an error term.

²⁹Indeed, in Section 5.5 we show that the decline in junior employment is driven by reduced hiring rather than increased separations.

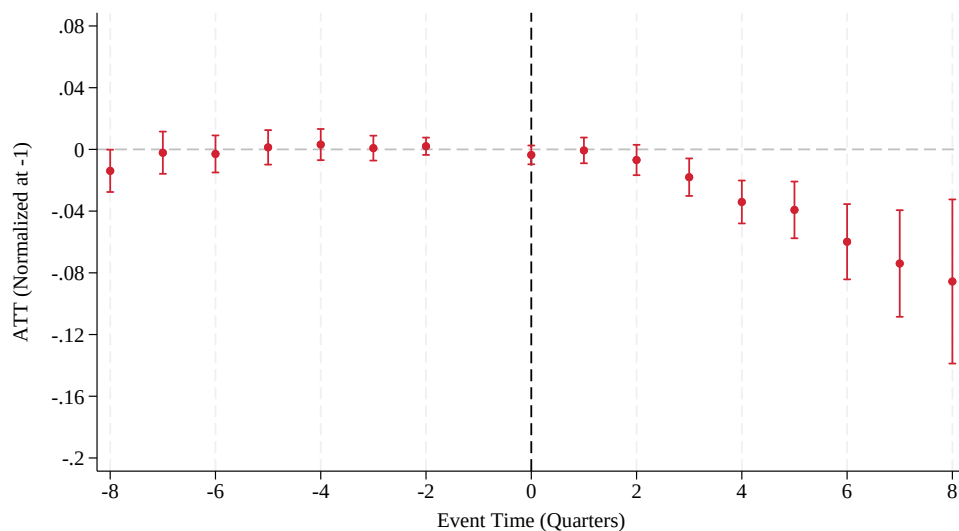


Figure 7: Event Study

Notes: The graph presents the estimated coefficients β_j from Equation 13 using the method of Callaway and Sant’Anna (2021) for staggered adoption. Firms with zero recorded employment in 2021Q1—eight quarters before the diffusion of GenAI—are excluded (2.3 percent of firms). Standard errors are clustered at the firm level.

Figure 7 reports the results. The coefficients remain flat during the pre-adoption period, supporting the parallel-trends assumption. Approximately two quarters after adoption, employment begins to decline, reaching an 8 percent reduction after eight quarters. The absence of significant pre-trends provides additional reassurance that these post-adoption declines are not driven by confounding shocks. Appendix A.27 repeats this exercise for senior employment and shows no comparable post-adoption decline.

5.4 Addressing Endogeneity Concerns

This section examines potential endogeneity concerns that could confound our interpretation of the estimated effects of GenAI adoption on junior employment. The central worry is that some other shock coinciding with the diffusion of GenAI—rather than GenAI itself—differentially reduced demand for junior workers at adopting firms. We consider the most plausible candidates: monetary tightening, the post-COVID hiring cycle, the normalization of remote work, and supply-side shocks to junior labor. In each case we explain why, although these factors are important for broader labor-market trends, they